

ERA5-Augmented Ensemble Learning for Multi-Horizon Tropical Cyclone Track Prediction over the Bay of Bengal: A Leakage-Aware Benchmark

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Abstract

Accurate short-to-medium-range cyclone track forecasting supports disaster preparedness over the Bay of Bengal, one of the world’s most cyclone-exposed basins. We benchmark eight track-prediction systems—Subset-Expert Context-Aware Ensemble (SECE v2 Phase 3) and seven baselines—at 3, 12, 24, and 48-hour lead times on North Indian Ocean IBTrACS storms (genesis 1940–2024). After quality control, 852 storms and 27,728 three-hourly origins define the modeling table; 583 storms (15,605 samples) support multi-horizon targets. Forty-nine track/physics inputs are augmented with 14 ERA5 point reanalysis features (winds at 850/500/200 hPa, mean sea level pressure, SST, steering flow, vertical shear). We identify and correct a subtle test-set leakage mode in which cross-run architecture decisions were guided by repeated inspection of test performance, and adopt a strict development versus held-out seed protocol (nine seeds each; held-out evaluated once). On pooled held-out tests (442 unique storm IDs), SECE significantly outperforms all seven baselines at 3 and 12 h (paired Wilcoxon on per-storm medians, Bonferroni correction, $p < 0.0071$). At 24 and 48 h, SECE is statistically indistinguishable from the strongest tree ensembles while remaining significantly better than CatBoost and, at 24 h, CB+MotionNN. Controlled ablations show that a dedicated atmospheric “environ” expert adds value over track-only features; spatial-patch ERA5 features were tested and not adopted. CNN-GRU and BLSTM were not re-evaluated on the locked held-out confirm.

1 Introduction

Operational numerical weather prediction (NWP) remains the reference for cyclone track guidance, but data-driven models trained on historical best tracks offer a lightweight complement where compute or assimilation capacity is limited [3,4]. For the Bay of Bengal rim—including Bangladesh, one of the most cyclone-vulnerable coasts—reliable 3–48 h track guidance feeds evacuation timing, port closures, and surge planning [1,2,28].

Despite growing interest in machine learning for tropical cyclones, three gaps persist for the North Indian Ocean: (i) narrow architectural benchmarking; (ii) inconsistent multi-horizon evaluation; (iii) unclear incremental value of atmospheric reanalysis beyond kinematics [10,16,17]. A fourth methodological gap is equally important: ensemble studies often iterate design while inspecting test-set rank tables, inflating apparent generalization unless confirm data are firewalled [19,20].

Contributions.

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- An eight-system, four-horizon benchmark under storm-wise splits with bootstrap CIs and pooled significance testing (442 unique test storm IDs across nine held-out seeds).
- Subset-Expert Context-Aware Ensemble (SECE v2 Phase 3), combining subset-specific tree experts, a dedicated ERA5 environ channel, champion baselines, and validation-only fusion routing (Phase 3).
- Controlled ablations (track-only, point ERA5, environ expert, $5^\circ/3^\circ$ spatial patches) with transparent negative results.
- A reproducible artifact trail aligned with the locked “NEW WAY” pipeline (datasets, held-out predictions, writeup tables).

2 Related work

Statistical-dynamical aids and climatological–persistence baselines remain operational references [8, 9, 25]. Machine-learning track models include tree ensembles and deep sequence architectures [11, 12, 24]; on tabular, engineered storm–environment data, trees often match or exceed deep networks at comparable sample sizes [13, 23]. ERA5 and satellite-era reanalysis provide gridded environmental context at storm locations [5, 14, 15, 29]. Basin-specific BoB studies have emphasized intensity or single-lead track tasks more than multi-horizon, leakage-controlled ensemble benchmarks [7, 17].

3 Data and experimental design

3.1 Sources and domain

Storm tracks come from IBTrACS v4 (North Indian Ocean) [6, 18]. ERA5 reanalysis [5] on $30\text{--}5^\circ\text{N}$, $70\text{--}102^\circ\text{E}$ is joined at each 3-hourly track time (≈ 10.5 GB source NetCDF over 551 storm-months).

3.2 Quality control and counts

Genesis-year filtering 1940–2024 yields 852 storms and 27,728 QC 3-hourly rows. Multi-horizon labels require sufficient forward track length; 583 storms contribute 15,605 multi-horizon origins (12, 24, 48 h). Storm-wise 70:15:15 train/validation/test splits are redrawn per seed ($\approx 408/87/88$ storms per held-out seed). Significance analysis pools test predictions across nine held-out seeds [0, 1, 2, 7, 13, 99, 123, 2024, 2026]; 442 is the count of distinct storm IDs in that pooled test set.

3.3 Features

Track/physics (49): position lags, motion kinematics, seasonality, recurvature proxies, land interaction, and missingness flags [30]. **ERA5 point (14):** 850/500/200 hPa winds, MSLP, SST (with coastal imputation flag; $\approx 38\%$ of rows), steering (u, v), shear (u, v , magnitude). Spatial $5^\circ/3^\circ$ patch statistics were extracted for ablation but not locked.

3.4 Date window

Fixed-test comparison of genesis windows 1940–2024, 1979–2024, and 1990–2024 supported retaining 1940–2024: long-lead advantage persisted on a common modern test draw while short-lead gaps were largely compositional [7, 21].

4 Test-set leakage and validation protocol

Early development advanced architectural phases using test-set rank inspection across a single split. Base learners were fit without test labels, but the *architecture* was tuned with test feedback—a common but under-reported leakage mode [20]. We separated nine *development* seeds [3, 4, 5, 6, 8, 9, 10, 11, 14] (architecture lock only) from nine *held-out* seeds evaluated once after freeze. Development and held-out mean medians align closely (e.g., 48 h SECE $\approx$ 247 km in both tiers), supporting that Phase 3 was not overfit to a single confirm split.

Primary metric: median great-circle (Haversine) error in kilometers. Held-out confirm adds 10,000 storm-bootstrap 95% CIs and paired Wilcoxon tests versus SECE with Bonferroni threshold $\alpha/7 \approx 0.0071$.

5 Methods

5.1 Baselines

Seven comparators: Stacking Ensemble (RF+XGB+LGB with Ridge meta), Persistence Residual Cascade (PRC), Random Forest, LightGBM, XGBoost, CatBoost, and CB+MotionNN (CatBoost + 64-32-16 MLP on motion residuals). Uniform tree hyperparameters (300 estimators, depth 6, learning rate 0.01) isolate architectural differences.

5.2 SECE Phase 3

Horizon-specific subsets (3 h: position, motion, environ, full; 12–48 h: motion, physics, environ, full) each train four tree algorithms, fused by NNLS per subset. Subset-NNLS signals join seven champions in horizon-specific fusion candidates (degree-NNLS, km-NNLS, stacking-residual at 3 h). A validation-best router selects the lowest validation median-km candidate per seed; test predictions use the frozen choice without test peeking (Fig. 1).

5.3 Training details

Table ?? summarizes locked hyperparameters (uniform across tree baselines unless noted). LightGBM uses 63 leaves; XGBoost subsample/colsample 0.8; Stacking Ridge meta-learner trained on 150 CV meta-predictions.

Table 1: Locked tree and meta-learner hyperparameters.

Component	Setting
Tree estimators	300
Max depth	6 (CatBoost: depth 6)
Learning rate	0.01
LightGBM num_leaves	63
XGBoost subsample / colsample	0.8 / 0.8
Stacking Ridge meta	150 CV meta-samples
MotionNN (CB+MotionNN)	64-32-16, lr 0.01, max 100 iter

5.4 Systems not in held-out table

CNN-GRU and BLSTM were explored in development but were **not** re-trained in the locked held-out run; they must not be cited on the confirm table.

6 Results

6.1 Development ablations

Table 1 reports SECE rank-1 frequency (seeds with lowest test median among eight systems) and mean median km across development seeds. Point ERA5 reduced 48 h mean error from 250.26 to 246.80 km (8/9 seeds improved). The environ expert improved 24 h mean to 99.43 km (best balance among tested configs). Spatial patches increased 3–12 h rank-1 frequency but did not justify complexity at 24–48 h (negative result).

Table 2: Development-seed ablation: SECE rank-1 count / mean median km (nine dev seeds).

Configuration	3 h	12 h	24 h	48 h
Track-only (no ERA5)	5/9·4.84	3/9·35.48	3/9·100.48	0/9·250.26
+ Point ERA5	5/9·4.93	7/9·35.59	2/9·99.86	1/9·246.80
+ Environ expert (locked)	5/9·4.93	6/9·35.56	3/9·99.43	2/9·246.94
+ Spatial 5°	7/9·4.93	7/9·35.56	1/9·99.74	1/9·246.55
+ Spatial 3°	7/9·4.93	7/9·35.66	2/9·99.42	3/9·247.00
Held-out (locked environ)	6/9·4.90	6/9·35.63	2/9·101.21	1/9·246.99

6.2 Held-out confirm

Table 2 lists pooled medians and CIs for all eight evaluated systems (34,114 test samples at 3 h; 21,442 at 12–48 h). At 3 and 12 h, SECE beats every baseline with Bonferroni significance (Table 3). At 24 h, differences versus Stacking, PRC, LightGBM, and XGBoost are not significant after correction; SECE remains significantly better than CatBoost and CB+MotionNN. At 48 h, SECE ties leading ensembles within CI overlap; only CatBoost is significantly worse. SECE is never significantly *worse* than any baseline at any horizon.

Fig. 2 shows the 3 h held-out error–accuracy landscape. Fig. 3 visualizes 63-D feature structure (10,000 subsample; $KL \approx 1.51$).

6.3 Per-storm ERA5 diagnostic

Track-only versus ERA5 LightGBM on development seed 0 ($n = 88$ test storms): at 24 h, 43/88 storms improve (48.9%); at 48 h, 54/88 improve (61.4%, median ≈ 10.9 km), consistent with steering information gaining relative importance at longer lead [15, 22].

6.4 Case studies

Laila (2010), SID 2010137N10090. Largest seed-0 ERA5 diagnostic gain (+25 km at 24 h, +107 km at 48 h vs. track-only LGB). SECE 24 h per-storm error ≈ 111.3 km (locally better than Stacking ≈ 128.9 km); at 48 h XGBoost leads locally (≈ 205 vs. 236 km), consistent with global 48 h rankings.

Typical 1996 storm, SID 1996163N08088. Central-basin genesis ($\approx 8.4^\circ\text{N}$, 87.7°E); SECE 24 h median ≈ 107.2 km, near seed-0 aggregate 24 h median (≈ 106.4 km). Fig. 4 maps Laila trajectories at 3/12/24 h (48 h omitted for readability; values in project CSVs).

6.5 Feature importance (illustrative)

Gain-based importances from environ-expert LightGBM (dev seed 0 only): kinematics still dominate; among ERA5 fields, SST, 200 hPa wind, and MSLP show largest shares. Shear and steering components are collinear—rank order within that group is not interpreted mechanistically.

7 Discussion

The central empirical finding is horizon-dependent: SECE provides a statistically robust advantage at 3–12 h, where storm-level context (recurvature proxies, coastal proximity, steering) carries strong learnable signal, but does not maintain statistically distinguishable superiority at 24–48 h, where it instead achieves parity with—never significant inferiority to—the strongest baselines [4, 13, 22].

We interpret this through bias–variance on ensemble complexity: as target uncertainty grows with lead time, marginal value of additional routing diminishes and well-regularized stacking becomes competitive. This extends tabular tree-versus-deep learning results one level further: *within* tree ensembles, complexity should track the information content available at each horizon rather than being applied uniformly [23].

ERA5 atmospheric features provide genuine value, but the locus differs from where architectural advantage is clearest: per-storm LightGBM diagnostics show majority-of-storms ERA5 benefit concentrated at 48 h, while the environ expert improved ensemble-level 24 h balance in development ablations [15, 16, 29]. We treat these as complementary, not identical, sources of improvement.

Reporting the leakage correction is not ancillary: initial phase advancement guided by test ranks produced optimistic short- and long-lead separation that did not replicate on fresh held-out seeds; dev/held-out alignment after freeze is the primary evidence that Phase 3 generalizes [19, 20].

8 Limitations and future work

Pre-1979 wind sparsity remains a caveat despite track-focused targets. Point ERA5 omits full spatial fields; gridded encoders, physics-constrained motion models, and probabilistic tracks are natural extensions [11, 19, 26, 27]. Multi-seed SHAP/permutation attributions would strengthen mechanistic claims beyond single-seed gain plots in Section 6. Three architectural directions may narrow 24–48 h gaps without revisiting leakage: (i) physics-constrained motion heads; (ii) probabilistic position forecasts with proper scoring rules [19, 27]; (iii) joint attention over storm history and gridded ERA5 [11, 24, 26].

9 Conclusion

We provide a leakage-corrected, ERA5-augmented Bay of Bengal track benchmark with honest significance reporting: SECE wins short range, ties the best at long range among evaluated tree systems, and documents negative spatial-patch results for reproducibility.

Data and code availability

Processed tables and held-out artifacts reside in the project repository ([NEW WAY/](#)); figures regenerate via `scripts/generate_paper_figures.py`. Held-out evaluation wall time ≈ 9.4 h (CPU, 16 GB RAM workstation); development ablations 4.9–10.9 h per configuration. ERA5 acquisition via Copernicus CDS for 551 storm-months required tens of hours of queue time; training remained CPU-bound (GPU unused). Seed lists, split counts, and storm-ID pooling definitions are exported in `docs/paper_writeup_data/`.

Acknowledgments

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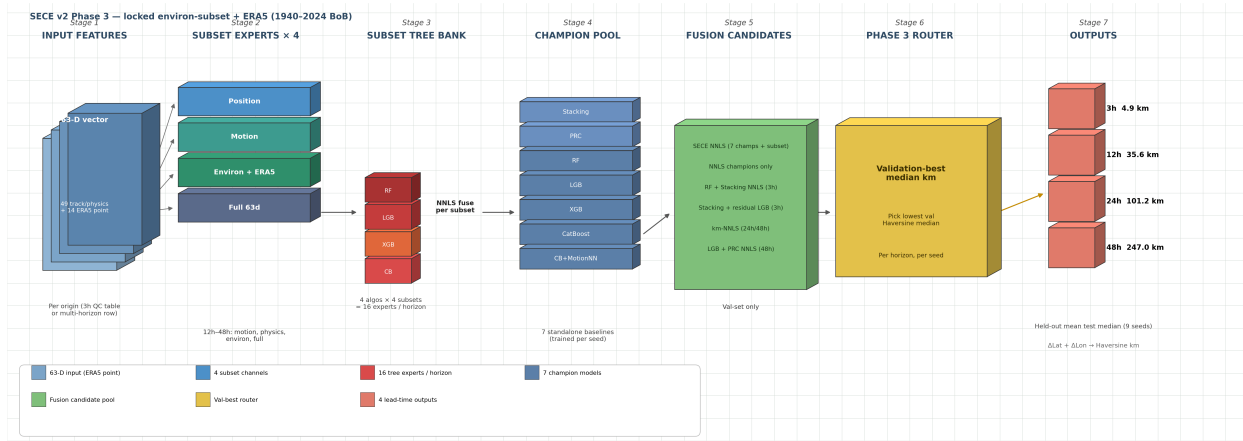


Figure 1: Locked SECE Phase 3 pipeline (63-D: 49 track/physics + 14 ERA5 point features).

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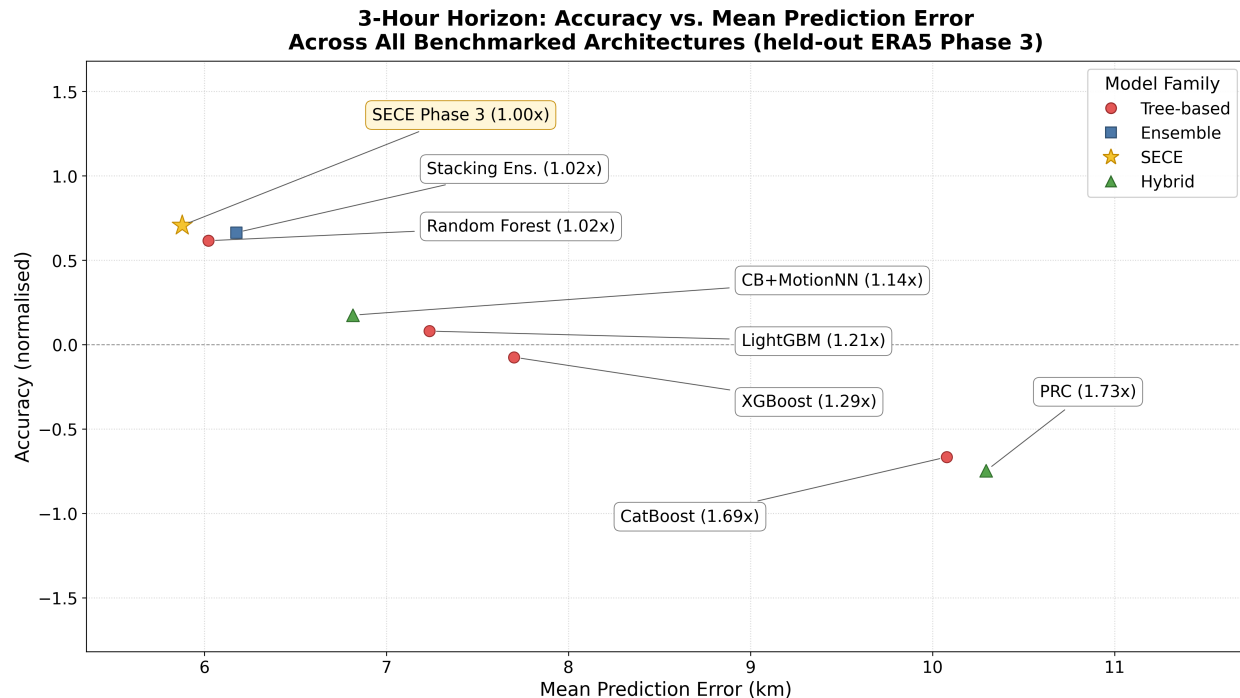


Figure 2: Three-hour held-out benchmark: eight systems (Phase 3 environ subset).

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t-SNE Projection of 63-Dimensional Training Feature Space
($n = 10,000$ training samples | KL = 1.509 | perplexity = 40)

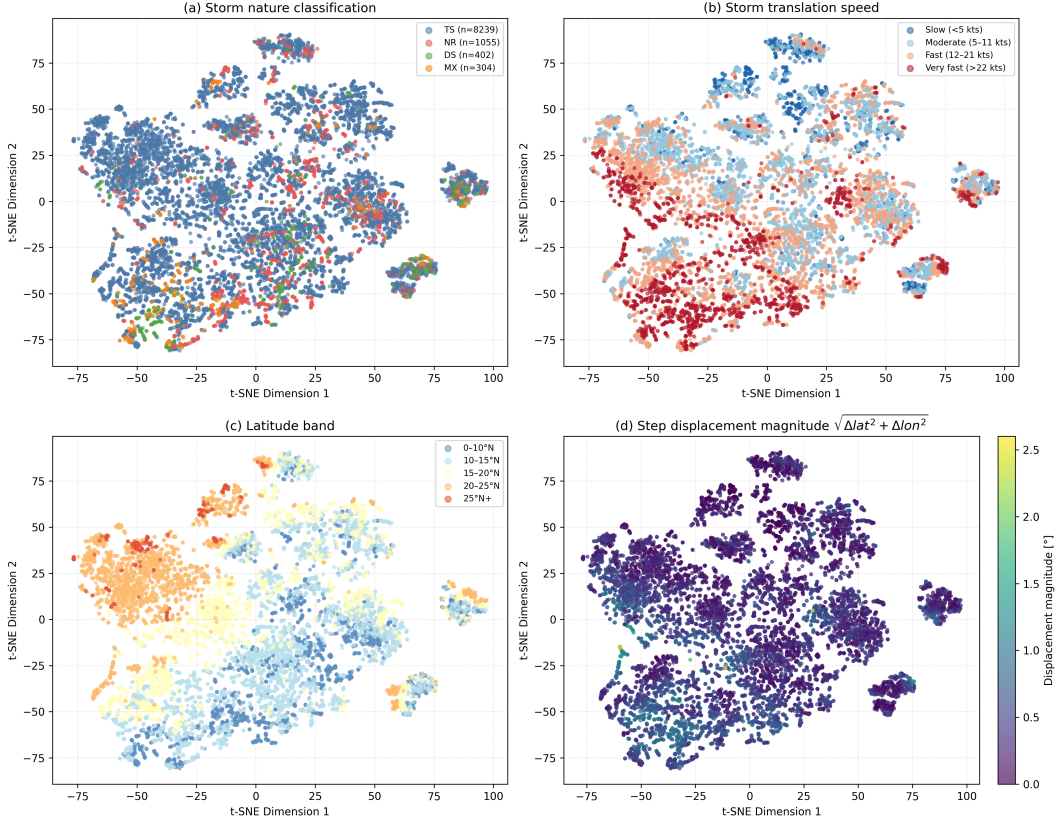


Figure 3: t-SNE of 63-D training features ($n = 10,000$ subsample).

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Table 3: Held-out test performance: median great-circle error (km) with 95% bootstrap CI. Nine held-out seeds pooled; 442 distinct storm IDs.

Horizon	System	Median	95% CI low	95% CI high
3 h	SECE v2 Phase 3	4.895	4.714	5.079
	Random Forest	4.927	4.734	5.125
	Stacking Ensemble	4.945	4.763	5.123
	CB+MotionNN	5.813	5.701	5.928
	LightGBM	6.026	5.889	6.168
	XGBoost	6.411	6.243	6.580
	CatBoost	8.479	8.284	8.691
	PRC	8.872	8.707	9.058
12 h	SECE v2 Phase 3	35.538	34.208	36.925
	Stacking Ensemble	36.146	34.744	37.701
	Random Forest	36.901	35.525	38.323
	LightGBM	36.928	35.498	38.338
	XGBoost	36.947	35.531	38.416
	PRC	37.177	35.704	38.527
	CB+MotionNN	37.413	36.047	38.949
	CatBoost	42.028	40.381	43.656
24 h	Stacking Ensemble	100.771	96.887	104.703
	XGBoost	101.029	96.932	105.286
	SECE v2 Phase 3	101.186	97.301	104.808
	LightGBM	101.632	97.778	105.589
	PRC	101.941	98.131	105.824
	Random Forest	103.511	99.652	107.437
	CB+MotionNN	104.034	100.239	108.074
	CatBoost	107.191	102.555	111.839
48 h	XGBoost	242.702	232.596	255.403
	PRC	243.229	231.650	255.158
	LightGBM	245.975	234.926	257.214
	Stacking Ensemble	245.977	235.938	257.183
	SECE v2 Phase 3	246.331	235.541	257.614
	CB+MotionNN	249.678	238.972	261.153
	Random Forest	251.305	240.752	262.324
	CatBoost	254.862	242.991	267.797

Table 4: Wilcoxon vs. SECE: Bonferroni-significant ($p < 0.0071$) pairs only (held-out).

Horizon	vs. baseline	Bonferroni sig.
3 h	All seven baselines	Yes (all)
12 h	All seven baselines	Yes (all)
24 h	CatBoost, CB+MotionNN	Yes
24 h	Stacking, PRC, LGB, XGB, RF	No
48 h	CatBoost	Yes
48 h	All others	No

Trajectory Validation — LAILA (2010): Held-out seed 0 vs IBTrACS

